

Chapter on Regression

Subject: Business Mathematics (Class 12 Management)

Syllabus: NEB, Nepal

1 Introduction: What is Regression and Why Do We Need It?

In our last chapter, we studied **correlation**, which tells us about the *strength* and *direction* of the relationship between two variables (like advertising expenses and sales). It answers the question, “Are these two related?”

Now, we move a step further with **Regression**. Regression helps us describe the *nature* of that relationship with an equation. It answers the question, “Okay, they are related, but *how* exactly? And can I use this relationship to make predictions?”

- **Correlation** is like knowing that studying more generally leads to better exam scores.
- **Regression** is like creating a formula that tells you *approximately what score you’ll get* if you study for a specific number of hours.

Introduction: Regression is a statistical method used to estimate or predict the value of one variable (the **dependent variable**) based on the known value of another variable (the **independent variable**).

Independent Variable (X): The cause, the predictor. This is the variable you control or know (e.g., Rainfall, Advertising budget, Study hours).

Dependent Variable (Y): The effect, the predicted value. This is the variable you want to estimate (e.g., Crop yield, Sales revenue, Exam score).

2 The Two Lines of Regression

In linear regression, we try to fit a straight line through our data points. This line is called the “line of best fit.” Because we have two variables, X and Y, we can create two different regression lines.

A. Regression Line of Y on X

This is the most common one. We use it to **predict Y when X is known**. Its equation looks very familiar, just like the equation of a straight line.

- **Equation:**

$$Y_c = a + b_{YX}X$$

where Y_c is the predicted value of Y, X is the given value of the independent variable, a is the Y-intercept, and b_{YX} is the **regression coefficient of Y on X**.

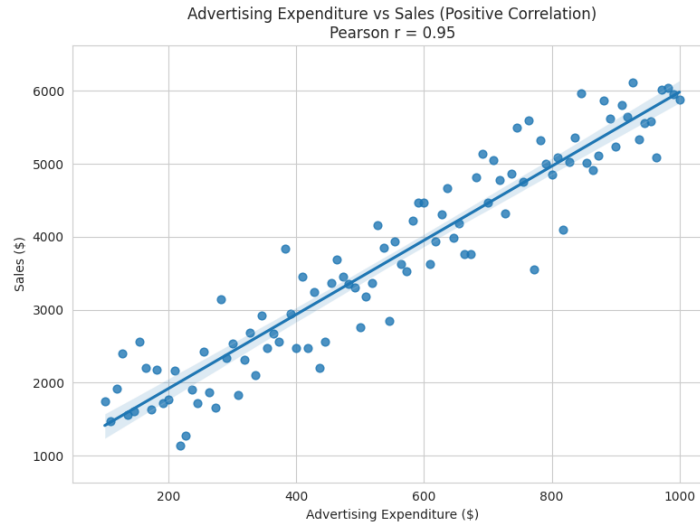


Figure 1: A scatter plot showing data points and the line of best fit.

B. Regression Line of X on Y

We use this line to **predict X when Y is known**.

- **Equation:**

$$X_c = c + b_{XY}Y$$

where X_c is the predicted value of X, Y is the given value of the dependent variable, c is the X-intercept, and b_{XY} is the **regression coefficient of X on Y**.

3 Finding the “Best Fit” - The Method of Least Squares

How do we decide where to draw our regression line? We use a method called the **Principle of Least Squares**. The “best” line is the one that is closest to all the points simultaneously. The least squares method finds the unique line that **minimizes the sum of the squared vertical distances** between each data point and the line itself. This ensures our predictions are, on average, as accurate as possible.

4 Calculating Regression Coefficients

To build our regression equations, we first need to calculate the coefficients (b_{YX} and b_{XY}).

Regression Coefficient of Y on X (b_{YX}):

$$b_{YX} = \frac{n(\sum XY) - (\sum X)(\sum Y)}{n(\sum X^2) - (\sum X)^2}$$

Regression Coefficient of X on Y (b_{XY}):

$$b_{XY} = \frac{n(\sum XY) - (\sum X)(\sum Y)}{n(\sum Y^2) - (\sum Y)^2}$$

Once you have the coefficients, you use these two equations to find the full regression lines:

- **For Y on X:** $Y - \bar{Y} = b_{YX}(X - \bar{X})$

- **For X on Y:** $X - \bar{X} = b_{XY}(Y - \bar{Y})$

Where $\bar{X}$ and $\bar{Y}$ are the means of X and Y, respectively.

5 Important Properties of Regression Coefficients

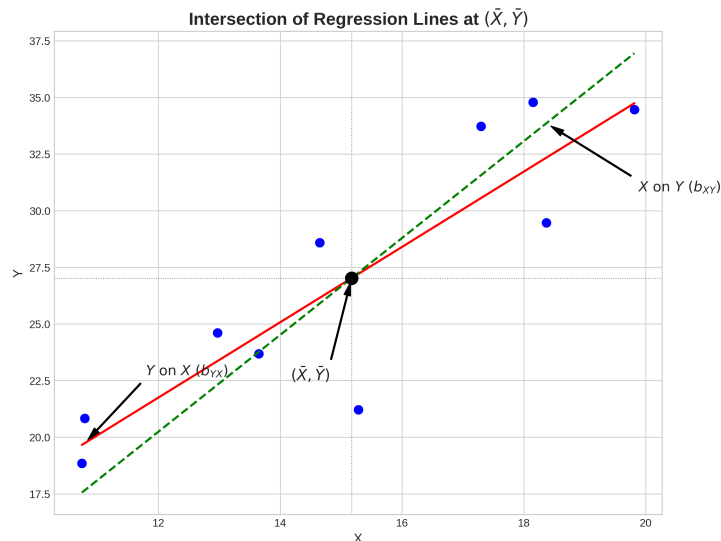
These are crucial for objective questions and for checking your answers!

1. **Same Sign:** Both b_{YX} and b_{XY} will always have the same sign (+ or -).
2. **Link to Correlation (r):** The correlation coefficient (r) is the geometric mean of the regression coefficients.

$$r = \pm \sqrt{b_{YX} \cdot b_{XY}}$$

The sign of 'r' will be the same as the sign of the coefficients.

3. **Magnitude:** If one of the regression coefficients is greater than 1, the other *must* be less than 1.
4. **Intersection:** The two regression lines (Y on X and X on Y) always intersect at the point $(\bar{X}, \bar{Y})$.



6 Solved Example (Typical NEB Question)

Question: From the data given below, find:

- a) The two regression equations.
- b) The most likely sales when advertising is Rs. 90,000.
- c) The correlation coefficient.

Advertising ('000 Rs) (X)	50	60	70	80	100
Sales ('000 Rs) (Y)	120	140	150	160	180

Solution:

Step 1: Create the Calculation Table ($n = 5$)

X	Y	XY	X²	Y²
50	120	6000	2500	14400
60	140	8400	3600	19600
70	150	10500	4900	22500
80	160	12800	6400	25600
100	180	18000	10000	32400
$\sum X=360$	$\sum Y=750$	$\sum XY=55700$	$\sum X^2=27400$	$\sum Y^2=114500$

Step 2: Calculate Means

$$\bar{X} = \frac{\sum X}{n} = \frac{360}{5} = 72$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{750}{5} = 150$$

Step 3: Calculate Regression Coefficients

$$b_{YX} = \frac{5(55700) - (360)(750)}{5(27400) - (360)^2} = \frac{278500 - 270000}{137000 - 129600} = \frac{8500}{7400} \approx 1.15$$

$$b_{XY} = \frac{5(55700) - (360)(750)}{5(114500) - (750)^2} = \frac{8500}{572500 - 562500} = \frac{8500}{10000} = 0.85$$

Step 4 (a): Formulate the Regression Equations

• **Y on X:**

$$Y - \bar{Y} = b_{YX}(X - \bar{X})$$

$$Y - 150 = 1.15(X - 72)$$

$$Y - 150 = 1.15X - 82.8$$

$$\mathbf{Y = 67.2 + 1.15X}$$

• **X on Y:**

$$X - \bar{X} = b_{XY}(Y - \bar{Y})$$

$$X - 72 = 0.85(Y - 150)$$

$$X - 72 = 0.85Y - 127.5$$

$$\mathbf{X = -55.5 + 0.85Y}$$

Step 5 (b): Make the Prediction

To predict sales (Y) when advertising (X) is 90, we use the Y on X equation:

$$Y = 67.2 + 1.15X$$

$$Y = 67.2 + 1.15(90)$$

$$Y = 67.2 + 103.5 = 170.7$$

The most likely sales are **Rs. 170,700**.

Step 6 (c): Calculate the Correlation Coefficient

$$r = \sqrt{b_{YX} \cdot b_{XY}} = \sqrt{1.15 \times 0.85} = \sqrt{0.9775} \approx \mathbf{0.988}$$

Since both coefficients are positive, 'r' is positive, indicating a very strong positive correlation.